
>

-
-
- `form` `label` `input` `button` `select` `textarea`
- **`Login.tsx`** `class` `className` `for` `htmlFor` `{ }`
-
- **`global.css`**
- **`Login.module.css`**

- `KosuList` `MainMenu` `Pagination`
-
-
-

>

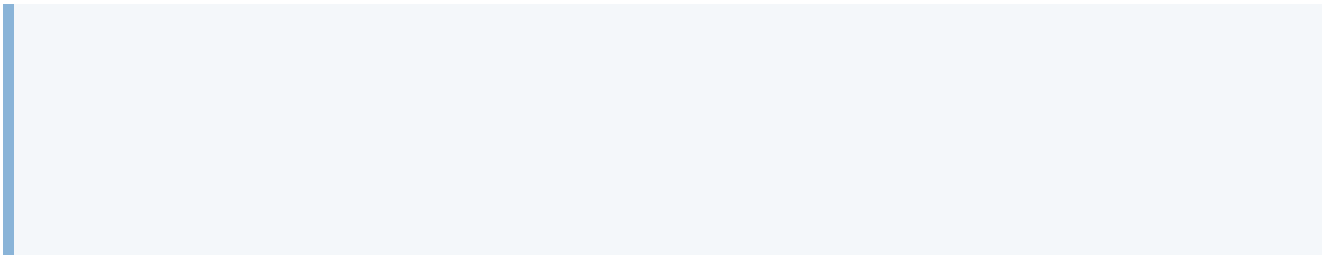
> `http://localhost:3000` `http://localhost:8000`

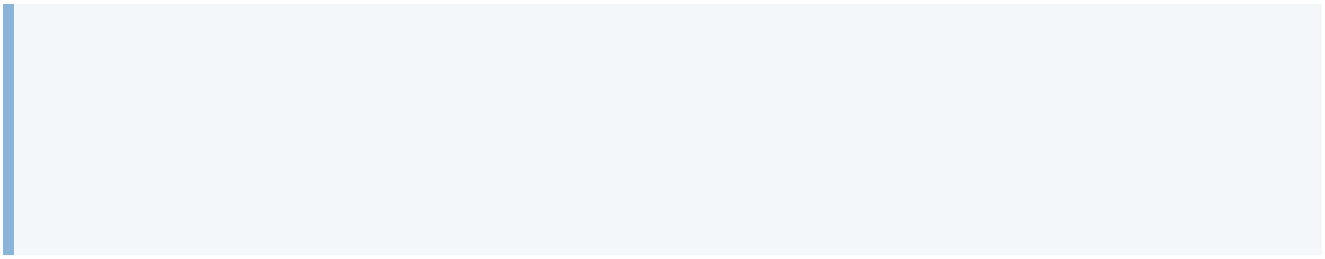
`http://localhost:3000`

```
1.                URL                                HTTP
2.              HTML                                 HTTP
3. HTML           DOM
4. CSS
5. DOM + CSS

6. JavaScript    DOM              3  5
```

>

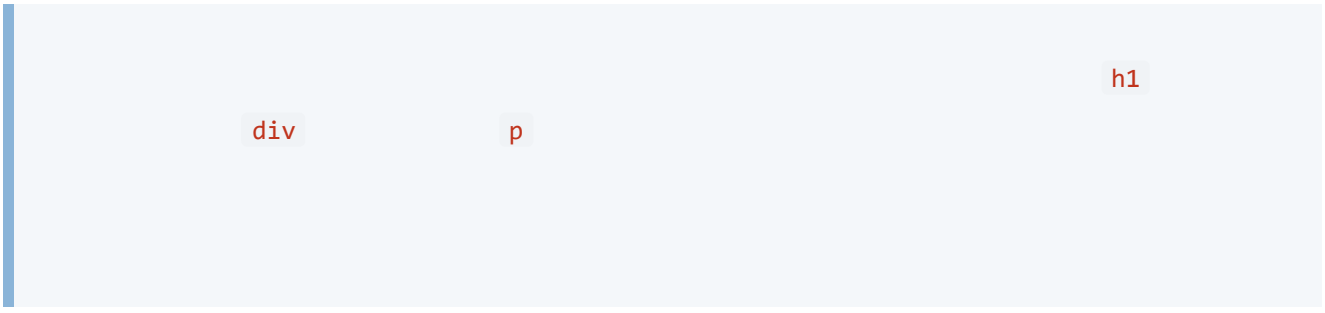




```
<div>
  <h1>        </h1>
  <p>    </p>
</div>
```

```
div
├─ h1         "        "
└─ p          "    "
```

`div` `h1` `p` `h1` `p`



`h1`

`div` `p`

			`.tsx`
			`styles/` `.css`
			`.tsx`

> `.tsx`

>

```
<p>                </p>
```

- `<p>`

-
- `</p>` `/`

> `<p>` `</p>`

```
<div>                <!--          -->
  <h1>      </h1>     <!--              -->
  <p>    </p>         <!--            -->
</div>               <!--               -->
```

> `<!-- ... -->`
>
> `<!--          -->`

> ⚠️ `<a><b></a>`
>
> `</b>` `<a><b></b></a>`

```
<img src="logo.png" alt="    ">   <!--                -->
<input type="text">               <!--                  -->
<br>                              <!--      -->
<hr>                              <!--        -->
```

>

⚠️ /

<img /> <input />

=" "

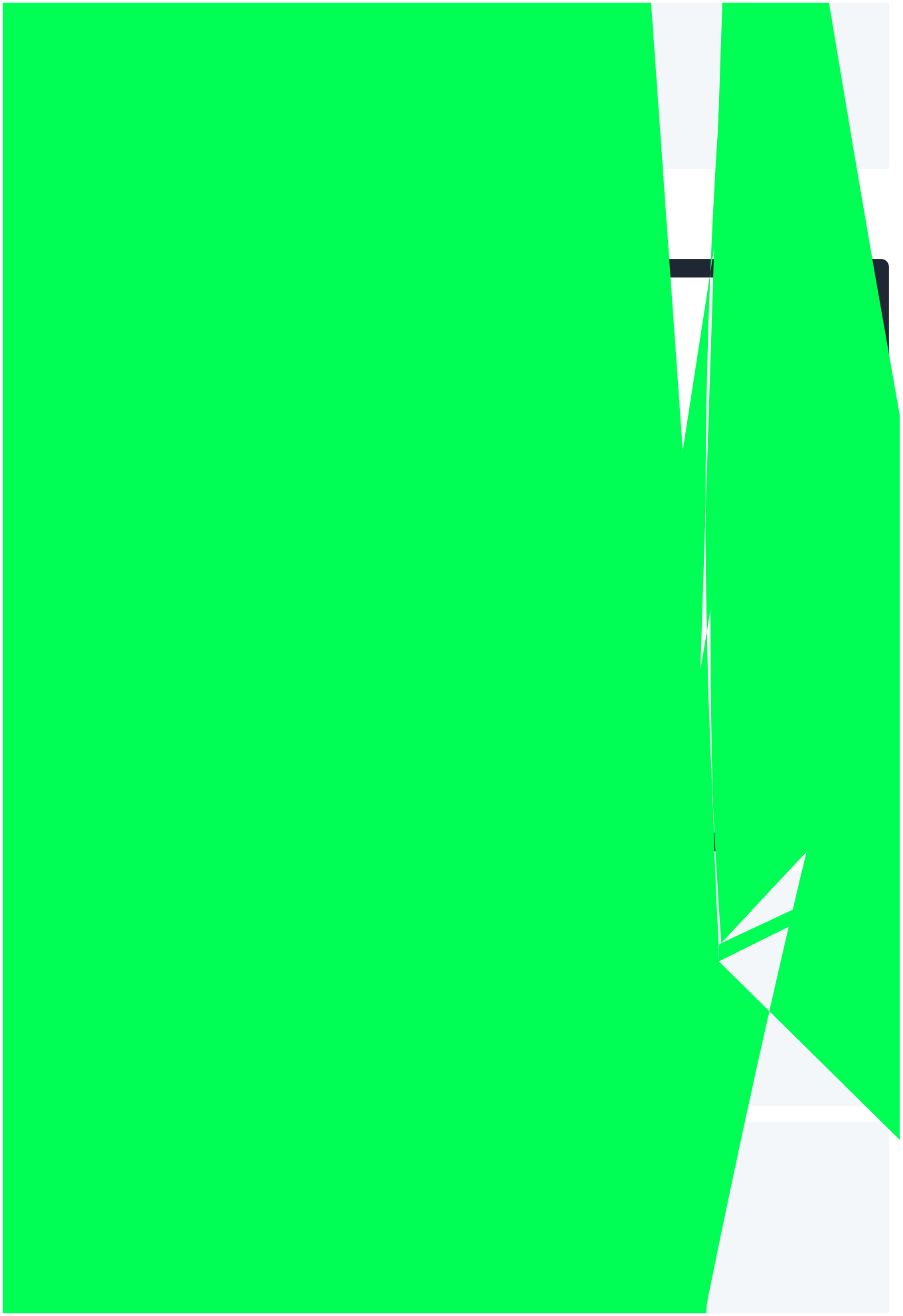
```
<a href="/help" target="_blank" class="link">      </a>
<!--  href=           target=         class=                  -->
```

id		id="numberInput"
class		class="blue_button"
href	<a>	href="/login"
src		src="logo.png"
alt		alt=" "
type		type="number"
value		value="123"
required		required
disabled		disabled
role		role="alert"

id

class

id # class .



> `div` `span`

>

>

```
<form>                                    <!--                              -->
  <label for="numberInput">          </label>   <!--              -->
  <input type="number" id="numberInput">       <!--        -->
  <button type="submit">        </button>       <!--            -->
</form>
```

- `<form>`
- `<label for="ID">` `for` `<input>` `id`
- `<button type="submit">`

`type`

`text`			
`number`			
`date`			
`checkbox`			
`password`			
`radio`			

> ⚠️ **`<input type="number">`**
>
> `Login.tsx` `Number(employee_no)`

```
<select>                        <!--                    -->
  <option value="P">P</option>   <!--       1 -->
  <option value="R">R</option>   <!--       2 -->
</select>

<textarea rows="4"></textarea>   <!--                              rows=
    -->
```

`ShopSelect` `InquirNew`

`<option value="...">` `value`

`frontend/src/MainPage/Login.tsx`

> - `class` `className`
> - `for` `htmlFor`
> - `{ }`

>

```
// ① React                          useState
import React, { useState } from "react";
// ②
import { Link, useNavigate } from "react-router-dom";
// ③                              4
import api from "../api/axios";
// ④
import axios from "axios";
// ⑤                         logo
import logo from "../img/TitleRogo.png";
// ⑥                 CSS  §12
import styles from "../styles/MainPage/Login.module.css";
```

- `import` `<img src={logo} />`
- `Login.module.css` `styles` `styles["login-wrapper"]`

> `login-wrapper`

```
        )}

        {/* ⑩              global.css           */}
        <button type="submit" className="blue_button">        </button>
      </div>
    </form>
    {/* ⑪                                §12    */}
    <Link to="/help" className={styles["help-button"]}></Link>
  </div>
);
```

- `styles["..."]` `Login.module.css` `styles["login-wrapper"]` `.login-wrapper`
- `<img />` `/>` `src={logo}` `{ }` `alt="  "`
- `<form>` `onSubmit` `onKeyDown` `onOO`
- `<h2>` `global.css` `h2`
- `htmlFor="numberInput"` `<input>` `id="numberInput"` `for` `htmlFor`
- `type="number"` `required`
- `{errorMessage && (<div role="alert">...)}` `errorMessage` `role="alert"`
- `className="blue_button"` `styles[...]`
- `<Link to="/help">` `help-button` `::after`

>

```
        {
            :   ;
            :   ;
}
```

```
/*                          p   */
p {
  /*          :   →                          ; */
  color: red;
  /*          16         */
  font-size: 16px;
}
```

- <p>
-
-
- ;

`p`		`<p>`
`.box`		`class="box"`
`#header`		`id="header"`
`.box p`		
`.box > p`		
`a:hover`		
`input[type="date"]`		
`div[role="alert"]`		
`*`		
`a, p`		
`th, td`		

>

`!important`

`style="..."`

`#id`

`.class` `[ ]`

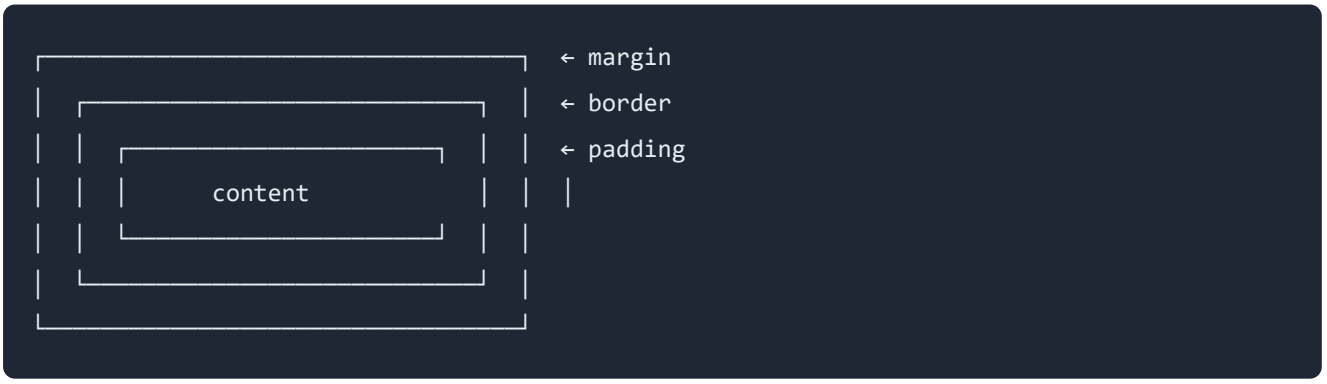
> ⚠️ **`!important`** `!important`
>
> `MainMenu.module.css` `display: none !important;`

`.css` `import`

`<div style="color:red">` `style={{color:'red'}}`

`<style>`

`global.css` `.module.css`



- `width` `height`
-
-
-

> `padding`
> `margin`

> `width`
> `width:200px; padding:10px`

`box-sizing: border-box` `width`
`width:200px`

```
/*              padding/border                                  §11.1        */
* { box-sizing: border-box; }
```

`global.css`

```
/*              10px */
margin: 10px;
/*     10px /     20px */
margin: 10px 20px;
/*   →  →  →               */
margin: 10px 20px 30px 40px;
/*     0      auto                        */
margin: 0 auto;
/*               */
margin-top: 8px;
```

> **`margin: 0 auto`** `auto`
> `login-logo`

> `form` `wrapper`

`px`		
`%`		
`rem`		
`em`		
`vw` `vh`		

> `vw`
> `vh` `100vh`
> `min-height: 100vh`

> **`clamp(最小, 推奨, 最大)`**

```
--heading1: clamp(1.5rem, 1.5rem + 3vw, 4rem);
/*     1.5rem                1.5rem+3vw      4rem */
```

calc(式) calc(100% - 60px)

min(値A, 値B) min(400px, 90%)

```
/*                    */
color: red;
/* 16     #RRGGBB  R   G   B     00  ff     */
color: #007bff;
/* 3          #00ff00              */
color: #0f0;
/*         0  255   */
color: rgb(0, 123, 255);
/*             (0=      1=      ) */
color: rgba(0, 0, 0, 0.1);
```

#007bff #0f0 #00ff00

#007bff #0f0

#0ff

---

display: flex

`display: flex`

```
.container {
  /*                                        */
  display: flex;
  /*           row=          , column=    */
  flex-direction: row;
  /*                          */
  justify-content: center;
  /*                           */
  align-items: center;
  /*                   */
  gap: 15px;
  /*                           */
  flex-wrap: wrap;
}
```

- **`justify-content`** `flex-start` `center` `flex-end` `space-between` `space-around`
- **`align-items`** `stretch` `center` `flex-start` `flex-end`

> `flex-direction: row` `justify-content` `align-items` `column`

`search-bar` `column` `KosuList` `search-bar` `row`

:

```
/*                      */
a:hover    { }
/*                                    */
input:focus{ }
/*              */
tr:nth-child(even) { }
/*              */
button:disabled { }
```

- :hover

```
```

```
.blue_button {
  background-color: #007bff;
  /*         0.3             */
  transition: background-color 0.3s ease;
}
.blue_button:hover {
  /*                   0.3                   */
  background-color: #0056b3;
}
```

- `transition:                              ;`
- `ease` `linear`
- `transition`

`@keyframes` `animation`

```css
/* fadeInUp                          */
@keyframes fadeInUp {
  /*             50px   */
  from { opacity: 0; transform: translateY(50px); }
  /*                      */
  to   { opacity: 1; transform: translateY(0); }
}
.login-wrapper {
  /*             1                                */
  animation: fadeInUp 1s ease-out;
}
```

- `opacity`
- `transform: translateY(50px)` `translateX`
- `from` `to`
-

`fadeInUp`

```css
/*         768px                    */
@media (max-width: 768px) {
  form {
    width: auto;
    /*                                              */
    min-width: unset;
  }
}
```

- `@media (    ) { ... }`
- `max-width: 768px`
- `unset`
- `global.css` `KosuList.module.css` `MainMenu.module.css` `Pagination.module.css`

`clamp()`

`frontend/src/styles/global.css`

```
/* *                                        */
* {
  /*                   0                          */
  margin: 0;
  /*                   0   */
  padding: 0;
  /* width  padding/border                  §5.1   */
  box-sizing: border-box;
}
```

`<h1>` `<p>` `<body>`

`box-sizing: border-box`

```
html, body {
  /*                                  */
  height: 100%;
  /*              Arial                    */
  font-family: 'Arial', sans-serif;
}
```

- `html, body`
- `height: 100%` `min-height: 100vh`
- `font-family` `sans-serif`

>

```
button,
input,
select,
textarea {
  /*                                                    */
  font-family: inherit;
  /*                      */
  font-size: 100%;
  margin: 0;
  padding: 0;
  /*                                    */
  border: none;
  /*                 */
  background: none;
  /*                           focus                */
  outline: none;
}
```



-





`min-height: 100vh` `overflow-x: auto`
`white-space: nowrap`

```
a {
  /*                    §11.4                    */
  text-decoration: none;
}

a:hover {
  /*                                    */
  text-decoration: underline;
}
```

`text-decoration` `none` `underline`

`<form>`

```css
form {
  /*        */
  background-color: #fff;
  padding: 0;
  /*               8px   */
  border-radius: 8px;
  /*                                  */
  box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
  /*       350px */
  min-width: 350px;
  /*       450px */
  max-width: 450px;
  /*     10px      auto             */
  margin: 10px auto;
  display: flex;
  /*                */
  flex-direction: column;
  box-sizing: border-box;
}

/*       768px           */
@media (max-width: 768px) {
  form {
    width: auto;
    /*                                      */
    min-width: unset;
  }
}
```

- `border-radius: 8px`
- `box-shadow: 0 4px 6px rgba(0,0,0,0.1)`

- `min-width: 350px; max-width: 450px`

- `margin: 10px auto`
- `min-width`

>

```
table {
  width: auto;
  /*                    1                    */
  border-collapse: collapse;
  margin-top: 10px;
  /*                      */
  white-space: nowrap;
}

th,
td {
  /*                */
  border: 1px solid #ddd;
  /*              */
  padding: 10px;
  /*          */
  text-align: center;
}

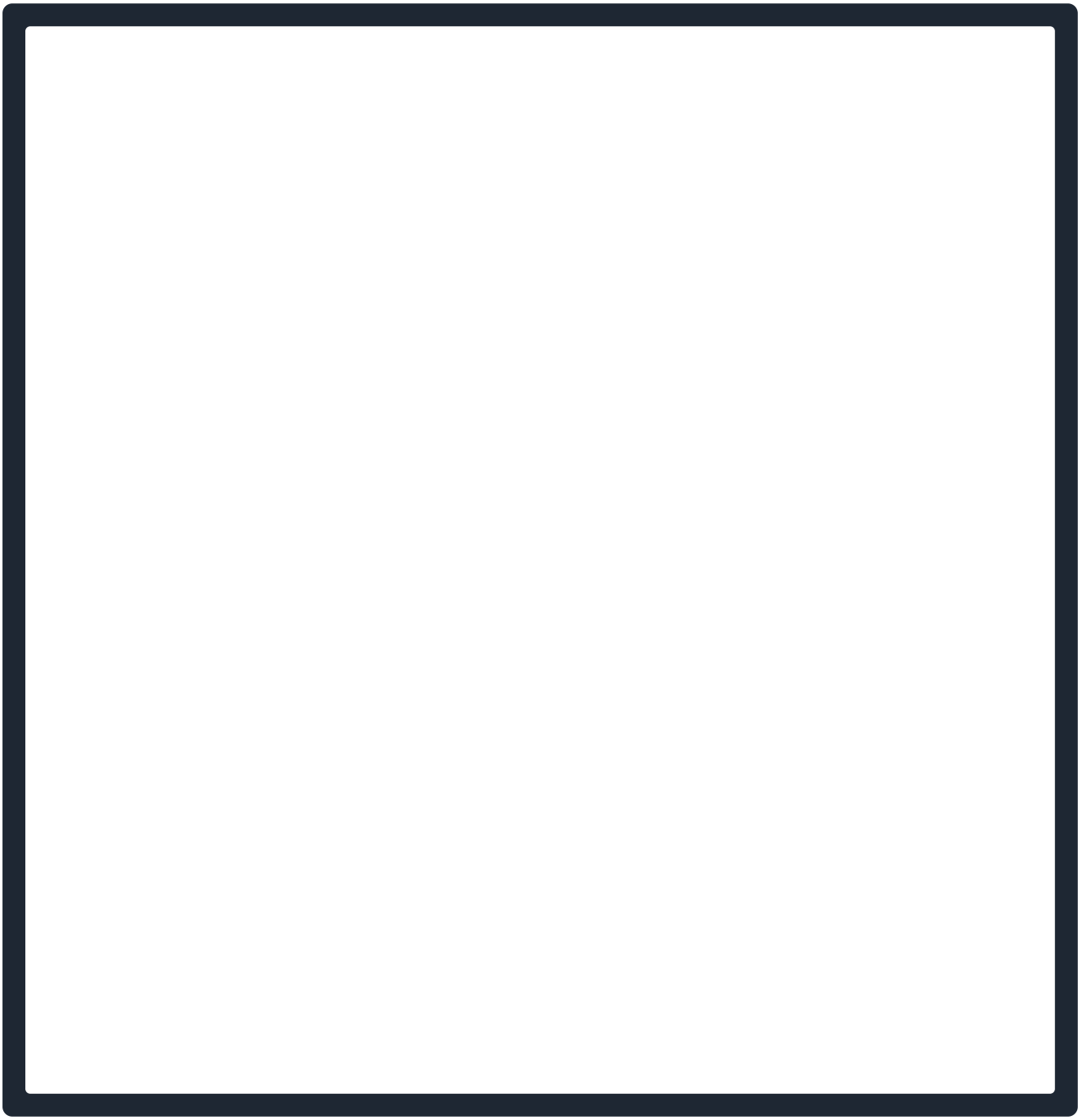
th {
  /*                      */
  color: white;
  font-weight: bold;
  text-shadow: -0.5px -0.5px 0 #555, 0.5px -0.5px 0 #555,
               /*       4                                */
               -0.5px 0.5px 0 #555, 0.5px 0.5px 0 #555;
}
```

- `border-collapse: collapse`

- `th, td`
- `th` `text-shadow`

`th-collar`

`KosuList`



- `writing-mode: horizontal-tb`
- `white-space: nowrap`
- `:hover` `transition`

`blue_button`	`#007bff`	`#0056b3`	
`yellow_button`	`#ffd700`	`#ffa500`	
`green_button`	`#0f0`	`#32cd32`	
`light_blue_button`	`#0ff`	`#0af`	
`orange_button`	`#f50`	`#f10`	
`pink_button`	`#ff8bc5`	`#ff1493`	
`gray_button`	`#656565`	`#3a3a3a`	

> `background-color`
>
> `:hover` `background-color`
>
> `.module.css`

> `role="alert"` `<div>`
>
> ⚠️ `Login.tsx`

```
/* role       alert    div                          */
div[role="alert"] {
  /*            */
  background-color: #f8d7da;
  /*            */
  color: #721c24;
  /*          */
  border: 1px solid #f5c6cb;
  border-radius: 4px;
  padding: 10px;
  /*   0      auto(    )    16px */
  margin: 0 auto 16px auto;
  font-size: 0.9rem;
  /*                               */
  white-space: pre-wrap;
  /*                     */
  overflow-wrap: break-word;
  /*                                  */
  word-break: break-all;
  /*         90%     */
  max-width: 90vw;
  box-sizing: border-box;
}

/*                          */
div[role="alert"]::before {
  /* ⚠️                      §8.2             */
  content: "⚠️ ";
  font-size: 1.1rem;
}
```

- `div[role="alert"]` `role` `alert` `Login.tsx` `<div role="alert">`

- `white-space: pre-wrap`
- `word-break: break-all`
- `::before` ⚠️



```
nav {
  display: flex;
  /*                          */
  justify-content: center;
  /*               15px */
  gap: 15px;
  margin-bottom: 20px;
}

nav a {
  background: none;
  padding: 0;
  border-radius: 0;
  /*                             */
  text-decoration: underline;
  font-weight: normal;
  transition: color 0.3s ease;
}
```

- `nav`
- `nav a` `text-decoration: underline` `transition: color 0.3s` `nav` `kosu-nav`

> `<input type="date">`

```css
input[type=date]::-webkit-calendar-picker-indicator {
  /*                                */
  position: absolute;
  /*                  */
  width: 100%;
  height: 100%;
  /*                                          */
  opacity: 0;
}

input[type="date"]::-webkit-inner-spin-button {
  /*                  */
  -webkit-appearance: none;
}

input[type="date"]::-webkit-clear-button {
  /*                    */
  -webkit-appearance: none;
}

input[type="date"] {
  /*             (absolute)               */
  position: relative;
}

.MuiPickersInputBase-root .css-15ni0jc {
  position: absolute;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  opacity: 0;
  /*                  */
  z-index: 1;
  cursor: pointer;
}
```

- `-webkit-`
- `opacity: 0`
- `position: relative` `position: absolute`

- `.MuiPickersInputBase-root .css-15ni0jc`

  `KosuList`

> `z-index: 1`

`global.css`

`frontend/src/styles/MainPage/Login.module.css`

`global.css`

> `<div>`

```
/*                                §9.2   */
@keyframes fadeInUp {
  from {
    /*            */
    opacity: 0;
    /*        50px                    */
    transform: translateY(50px);
  }
  to {
    /*              */
    opacity: 1;
    /*                  */
    transform: translateY(0);
  }
}

.login-wrapper {
  /*           1                   */
  animation: fadeInUp 1s ease-out;
  width: 100%;
  max-height: 100%;
  /*                  */
  max-width: 100vw;
  /*                        */
  overflow-x: auto;
  overflow-y: auto;
  /* iOS                          */
  -webkit-overflow-scrolling: touch;
  /*            */
  margin: 0 auto;
}
```

-

```css
.search-bar {
  display: flex;
  /*               →    →               */
  flex-direction: column;
  justify-content: center;
  /*          */
  align-items: center;
  /*             15px */
  gap: 15px;
  margin: 20px 0;
  flex-wrap: wrap;
}
```

⚠️ `search-bar` `flex-direction: column` `KosuList` `search-bar` `row`

```css
.login-logo {
  /*     400px                    90%      min       */
  max-width: min(400px, 90%);
  width: auto;
  /*              */
  height: auto;
  /*        */
  margin: 0 auto;
  /*     40px                              */
  margin-bottom: 40px;
  /*                          */
  object-fit: contain;
}
```

- `min(400px, 90%)`

- `object-fit: contain`

> `outline: none`

```
.search-bar input {
  /*                    8px    10px   */
  padding: 8px 10px;
  /*          */
  border: 1px solid #999;
  border-radius: 4px;
  font-size: 1rem;
  /*           100%    60px            calc       */
  width: calc(100% - 60px);
}

/*                      §8.1              */
.search-bar input:focus {
  /*          */
  border-color: #08f;
  outline: none;
  /*                                */
  box-shadow: 0 0 3px #555;
}
```

- `calc(100% - 60px)`
- `:focus` `#08f` `box-shadow`

`Login.tsx` `<Link className={styles["help-button"]}>`

>

```css
.help-button {
  display: flex;
  align-items: center;
  /*       ?           */
  justify-content: center;
  width: 20px;
  height: 20px;
  /*        */
  background-color: #ffffff;
  /*        */
  border: 2px solid #007bff;
  /* 45                              */
  transform: rotate(45deg);
  /*                                          */
  position: fixed;
  /*           30px */
  bottom: 30px;
  /*           30px */
  right: 30px;
  text-decoration: none;
  cursor: pointer;
  /*                          */
  opacity: 0;
  /* 30                     */
  animation: fadeIn 0s linear 30.0s forwards;
}

@keyframes fadeIn {
  to {
    /*     →       */
    opacity: 1;
  }
}

/*                                     */
.help-button:hover {
  /*                                           */
  background-color: #007bff;
}

/*                   ?          */
.help-button::after {
  content: "?";
  /*       45                             -45                */
  transform: rotate(-45deg);
  color: #007bff;
  font-weight: bold;
  font-size: 20px;
```

```
}

/*                      */
.help-button:hover::after {
  /*                ?         */
  color: #ffffff;
}
```

- **`transform: rotate(45deg)`** `::after` `?` **`rotate(-45deg)`**

- **`position: fixed; bottom: 30px; right: 30px;`** `fixed` `absolute`
- **`animation: fadeIn 0s linear 30.0s forwards`** `fadeIn` `linear` `forwards`
  - `0s`
  - `30.0s`
  - `forwards`
  -

> `help-button` `blue_button`

`Login.module.css`

`global.css` `Login.module.css`

`frontend/src/styles/KosuPage/KosuList.module.css`

```
.kosu-list-wrapper {
  /* Login                                                */
  animation: fadeInUp 1s ease-out;
  display: flex;
  /*        */
  flex-direction: column;
  justify-content: top;
  /*          */
  align-items: center;
  height: 100vh;
  width: 100%;
  margin: 0 auto;
  -webkit-overflow-scrolling: touch;
}

.th-collar {
  /*                          global.css  th                              */
  background-color: #0ff;
  text-shadow: -0.5px -0.5px 0 #555, 0.5px -0.5px 0 #555, -0.5px 0.5px 0 #555, 0.5px
0.5px 0 #555;
}

.status-ok {
  /*   OK                 */
  color: blue;
  font-weight: bold;
}

.status-ng {
  /*   NG                 */
  color: red;
  font-weight: bold;
}
```

- `th-collar` `global.css` `th`
  `<th className={\`

- `status-ok` `status-ng`

```
.search-bar {
  display: flex;
  /*         Login  search-bar  column             */
  flex-direction: row;
  justify-content: center;
  align-items: center;
  gap: 15px;
  margin: 20px 0;
  /*                          */
  flex-wrap: wrap;
}

@media (max-width: 768px) {
  .search-bar {
    /*                               */
    flex-direction: column;
    gap: 10px;
  }
  .kosu-nav a {
    /*                  */
    font-size: 0.8rem;
  }
}
```

> `row` `column`

`frontend/src/MainPage/MainMenu.tsx` `<Link>`

```
{/*                          */}
<Link to="/kosu-menu" className={styles["kosu-menu-button"]}>
  {/*              */}
  <img src={Note} alt="            " className={styles["icon"]} />
  {/*              */}
      MENU
</Link>
```

> <Link>

```
.kosu-menu-button {
  width: 100%;
  max-width: 400px;
  /*      */
  background-color: #0ff;
  color: #fff;
  /*                          §6.2   */
  font-size: clamp(1rem, 1rem + 3vw, 2rem);
  border-radius: 4px;
  cursor: pointer;
  transition: background-color 0.3s ease, box-shadow 0.3s ease;
  display: flex;
  align-items: center;
  /*                           */
  justify-content: space-between;
  padding: 10px 15px;
  position: relative;
}
.kosu-menu-button:hover {
  /*                  */
  background-color: #0af;
  /*                */
  box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
}
```

MainMenu

global.css

```
.kosu-menu-button::after,
.def-menu-button::after,
/* ...             ... */ {
  /*                     */
  content: "";
  height: 1em;
  flex: 0 0 auto;
  /*                                    */
  width: var(--icon-width, 1em);
}

@media (max-width: 768px) {
  .admin-menu-button {
    /*                                */
    display: none !important;
  }
}
```

- `::after` `justify-content: space-between`

- `display: none !important`
  `!important`

  `frontend/src/styles/Components/Pagination.module.css` `clip-path`

> `clip-path`

```css
.pagination {
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 15px;
  margin-top: 20px;
  white-space: nowrap;
}

.pagination .next-button {
  position: relative;
  width: 70px;
  height: 50px;
  /*     JS                 */
  background-color: var(--btn-bg-color, #fff);
  border: none;
  /*     →      →                */
  clip-path: polygon(0% 0%, 100% 50%, 0% 100%);
  cursor: pointer;
  transition: background-color 0.3s ease, box-shadow 0.3s ease;
  display: flex;
  align-items: center;
  justify-content: flex-start;
  padding-left: 10px;
  color: #FFF;
  font-weight: bold;
  text-shadow: -0.5px -0.5px 0 #555, 0.5px -0.5px 0 #555, -0.5px 0.5px 0 #555, 0.5px 0.5px 0 #555;
}

.pagination button:disabled {
  /*                      */
  background-color: #999;
  color: #666;
  /*                            */
  cursor: not-allowed;
}
```

- `clip-path: polygon(0% 0%, 100% 50%, 0% 100%)`
- `var(--btn-bg-color, #fff)` `, #fff`
- `:disabled`

`cursor: not-allowed` 🚫

> `Pagination`

`styles["login-wrapper"]`

> `.module.css`

`global.css`
`.search-bar`
`Login` `KosuList` `.search-bar`

```
// styles
import styles from "../styles/MainPage/Login.module.css";
// ...
// styles["login-wrapper"]
<div className={styles["login-wrapper"]}>
```

`login-wrapper` `Login_login-wrapper__a1B2c`
`KosuList` `search-bar` `KosuList_search-bar__x9Y8z` `Login`
`search-bar` `Login_search-bar__p3Q4r`

Login.tsx

```
// global
<button type="submit" className="blue_button">        </button>
// module  styles
<div className={styles["search-bar"]}>
```

- global.css className="blue_button"
- .module.css className={styles["..."]}

	global.css	.module.css
	className=" "	className={styles[" "]}

⚠️ className="login-wrapper" styles["login-wrapper"] styles["blue_button"]

-

`blue_button` `global.css` `.module.css`

`Login_login-wrapper__a1B2c` `__` `login-wrapper` `.module.css`

-
-

>

>

- `<label htmlFor="numberInput">`

- `<div role="alert">` `role="alert"`

- `alt="` `"`
  `alt=""`

> ⚠️ `htmlFor` `id`
> **`alt`** `outline: none`
> `:focus`

		`styles["  "]` `"  "`
	`.module.css`	
	`box-sizing`	`box-sizing: border-box`
	`row` `column`	`flex-direction` `justify-content` `align-items`
		`border-collapse: collapse`
	`:hover`	
		`height: auto`
`;`		
	`outline: none`	

`MainMenu` `Pagination` `clip-path`

- `CSS Modules` `<button className="blue_button">` `<div className={styles["search-bar"]}>`
- `htmlFor` `id` `role="alert"` `alt`

`onChange` `useState` `Number()` `{ && }`